

Software to Help You Select Inverter for Your Home



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Choosing the right ratings for the inverter and its battery to be used at home is not that

easy. There are many parameters that you need to know before buying an inverter, such as load requirement, size of the inverter, back-up time, size of battery, etc. Presented here is a

Software program

Python is a high-level programming language with rich syntax. There are lots of packages available for Python. We used wxpython package for graphical user interface (GUI).

First you need to install Python; we installed its 2.7.6 version. You can download it from the link <https://www.python.org/>. Then you need to install wxpython package. You can get wxPython3.0-win32-py27 for 32-bit Python 2.7 version or wxPython3.0-win64-py27 for 64-bit Python 2.7 version from www.wxpython.org/download.php.

To run the application, just right

File Help

INVERTER SELECTION FOR YOUR HOME

	watts	Hrs	
load 1	3	1	power factor 0.7
load 2	0	1	battery volt 12 V
load 3	0	1	backup time 3 Hrs
load 4	0	1	Enter
load 5	0	1	total power you need 0 watts
load 6	0	1	Required VA of inverter 0 VA
load 7	0	1	battery capacity 0 Ampere Hrs
load 8	0	1	
load 9	0	1	
load 10	0	1	
load 11	0	1	
load 12	0	1	
load 13	0	1	
load 14	0	1	
load 15	0	1	
load 16	0	1	
load 17	0	1	
load 18	0	1	
load 19	0	1	
load 20	0	1	

Note: default values must be load = 0 Watts and Hrs = 1, don't empty the text box

LOAD REFERENCE TABLE

DEVICE	WATTS
Coffee maker	900-1200
Ceiling fan	45-175
Window fan	55-250
Heater (portable)	750-1500
CPU(jawake/sleep)	120/30 or less
Monitor(jawake/sleep)	150/30 or less
Laptop	50-100

Fig. 1: Program output of inverter selection

program in Python that helps you select the right inverter based on some common parameters. This article also describes how to create a standalone application so that you can run the program without Python software, if needed.

```

setup(
    windows = [
        {
            "script": "inverterApplication.py",
            "other_resources": [(24,1,manifest)]
        },
    ],
)

```

Fig. 3: setup.py script

File Help

INVERTER SELECTION FOR YOUR HOME

	watts	Nos		
load 1	40	3	power factor	0.7
load 2	70	3	battery volt	12 V
load 3	120	1	backup time	3 Hrs
load 4	100	1	Enter	
load 5	0	1	total power you need	\$50.0 watts
load 6	0	1	Required VA of inverter	785.7142857 VA
load 7	0	1	battery capacity	127.5 Ampere Hrs
load 8	0	1		
load 9	0	1		
load 10	0	1		
load 11	0	1		
load 12	0	1		
load 13	0	1		
load 14	0	1		
load 15	0	1		
load 16	0	1		
load 17	0	1		
load 18	0	1		

Note: default values must be load = 0 Watts and Nos = 1.
don't empty the test box

DEVICE	WATTS
Coffee maker	900-1200
Celling fan	65-175
Window fan	55-250
Heater(portable)	750-1500
CPUs(make/sleep)	120/35 or less
Monitor(make/sleep)	150/35 or less
Laptop	50-100

Fig. 2: Example calculation

```
<asmv:windowsSettings
  xmlns="http://schemas.microsoft.com/BIM/2005/WindowsSettings">
  <applicationStructure>{E6A90A68-9848-4678-A081-E1875A036F58}
</asmv:windowsSettings>
</asmv:application>
<assemblyIdentity
  version="2.6.25.0"
  processorArchitecture="x86" replace with "x64"
  name="Dropbox"
  type="win32"
/>
<description></description>
<dependency>
  <dependentAssembly>
    <assemblyIdentity
      type="win32"
      name="Microsoft.Windows.VC90.CRT"
      version="9.0.30729.1"
      processorArchitecture="x86" replace with "x64"
      publicKeyToken="9596b6417eabd662"
    />
  </dependentAssembly>
</dependency>
<dependency>
  <dependentAssembly>
    <assemblyIdentity
      type="win32"
      name="Microsoft.Windows.Common-Controls"
      version="6.0.0.0"
      processorArchitecture="x86" replace with "x64"
      publicKeyToken="659586410eafa0c6"
      language="en"
    />
  </dependentAssembly>
</dependency>
</asmv:application>
```

Fig. 4: *setup.py* for 64-bit machine